VoiceOps 2026: ZentroMart customer-support agent

Team: Kaizeen · Repository: github.com/ChiragSharma-DEV/ZentoVoiceOPS · Data as-of date: 2026-10-20 · LLM: gemini-3.5-flash-lite, temperature 0 · Voice: Gnani gnani-vachana 0.8.2

1. Summary

We built a ZentroMart support agent in four steps. Every number here comes from a run output in the repo. Step 1, a prompt-only baseline, got 15 of 52 decisions and 8 of 52 must-include facts right. Step 2 adds retrieval over the knowledge base. The frozen configuration scores 26/26 decisions on the holdout text split and 25/26 must-include facts on the holdout ASR split. Step 3 is a conversation agent with identity verification, two tools, hardcoded escalation and handover. It passes 81 of 96 turns on 23 scenarios; the earlier version passed 79 on three runs. Step 4 runs speech in and out. Two of six voice clips pass. Three fail: speech recognition dropped key terms in two, and one reference omits what the caller said. Known gaps: return deadlines are not computed, one Step 3 turn regressed when an order fact changed retrieval (reconstructed, not directly recorded), and Step 3 scenarios were also used for development.

2. What was built, by step

- **Step 1 (baseline).** One prompt per question to gemini-3.5-flash-lite over REST at temperature 0, with a last-JSON parser. Scored by `src/scoring.py` `score_row`, which checks must-include and must-not facts against the reply only.
- **Step 2 (RAG).** Hybrid retrieval over `kb/*.md` (local embeddings and faiss). The answer prompt cites KB section IDs. Refund dates are computed in code. ASR normalisation (Fix 5) applies to voice input only. Fixes 1, 4 and 6 are off in the frozen configuration.
- **Step 3 (conversation).** `ConversationAgent` in `src/conversation.py`. Fixed flows cover greeting, identity verification (IDV-1, IDV-2), order lookup, ticket creation and handover. Policy questions go to the Step 2 pipeline. Escalation rules are hardcoded from the KB. Section citations are checked, and unknown IDs are dropped and logged.
- **Step 4 (voice).** Gnani STT (en-IN), then the agent with `is_asr=True`, then Gnani TTS (voice Kaveri, model timbre-v2.5). Hinglish replies use language "auto", because the SDK rejects "hi-en".

3. Step 1 versus Step 2

Run	Rows	Decision correct	Must-include correct	Failing IDs
Step 1, prompt only	52	15/52 (28.8%)	8/52 (15.4%)	37 decision failures
Step 2, DEV ASR (frozen)	26	23/26 (88.5%)	22/26 (84.6%)	Decision: PQ0020, PQ0030, PQ0046. Must-include: PQ0020, PQ0024, PQ0026, PQ0028
Step 2, HOLDOUT text (frozen)	26	26/26 (100%)	24/26 (92.3%)	Must-include: PQ0005, PQ0010
Step 2, HOLDOUT ASR (frozen)	26	26/26 (100%)	25/26 (96.2%)	Must-include: PQ0027

The frozen configuration was chosen after Phases 2B and 2D scored all 52 rows, including HOLDOUT. HOLDOUT is a development split, not an unseen test. Fix 5 changed six DEV ASR queries: PQ0002, PQ0026, PQ0030, PQ0032, PQ0041 and PQ0046. PQ0028 fails its must-include check in the frozen DEV run, and Fix 5 did not change its query.

4. Step 3 results

Scenarios PC002 to PC024 (23 scenarios, 96 turns). PC001 has no rows in the dataset. Each turn is scored on decision, must-include and must-not. Must-not violations: 0 in every run. Exceptions: 0.

Run	Pass	Decision failures	Must-include failures	Tool-call failures
v1 (first code)	79/96	6	16	2
v1 replicate 1	79/96	6	16	2
v1 replicate 2	79/96	6	17	2
v2 (final code)	**81/96**	4	15	0

The final v2 re-run after packaging gives the same totals and failing set. Reply wording changed on 12 turns at temperature 0.

v2 changes, each driven by a v1 failure:

- A human-agent request hands over at once, before verification (ESC-1 item 6). Fixes PC005 T4.
- Rule 5 read field names that do not exist in orders.csv, so it never fired. It now reads the dataset's return, cancellation and RTO refund fields. PC010 T4 now escalates, which matches the expected decision, but still misses the RRN and ticket wording.
- `shipped_on` added to the verified-order facts. Fixes PC013 T5 and PC014 T5.
- Policy questions with no order cue are answered without verification (IDV-1). Claims and complaints still go to the identity gate.

Regression. PC013 T4 now fails in both v2 runs, after `shipped_on` was added to the verified-order facts. Its v1 reply gave the one-coupon rule. Reconstructing the v1 and v2 queries from the recorded customer input and the order block (the composed query is not stored in the transcripts), adding the `shipped_on` line removes both coupon chunks, CPN-1 and CPN-2, from the top 10. The v2 reply then says it cannot find the answer. The mechanism is supported but not directly recorded.

Remaining 15 failures: four refusals answered instead of refused (PC015, PC016, PC019, PC020 T4); two "cannot find" non-answers (PC004 T4, PC012 T4); no computed return deadline (PC003 T4); a wrong timeline (PC003 T5); missing timeline ranges, since Fix 6 is off (PC004 T5); order facts not given after verification (PC011 T3, PC012 T3); a missing QC stage (PC009 T4); a missing "3" (PC011 T4); the RRN gap (PC010 T4); and the regression above.

The Step 3 scenarios informed the v1 fixes, so v2 is not an unseen-data estimate. Handover records are in `tickets_created.csv` in each batch folder.

5. Voice results

Six clips, run through Gnani STT and TTS with 12 Gnani calls per run. Zero STT or TTS errors. The voice runs made no Gemini calls. Clip 03 is replayed in its Step 3 context. Clips 04 to 06 are single turns, and clip 01 is a verification turn. STT output is compared with the reference text.

Clip	Input	Outcome
01 verification	Spoken order ID and a delivery question	**Pass.** Spoken digits parsed to 69520867, verified, question answered.
02 damaged item	Damage and a compensation demand	Not scored: no valid expected row. The claim gate asked for identity, which is correct.
03 human agent	Asks to speak to a human	**Pass.** Escalates, matching PC006 T4 in the Step 3 outputs.
04 return policy	Return policy question	**Fail.** STT produced "lutane"; no return cue matched; the agent asked for clarification.
05 warranty claim	Warranty claim question	**Fail.** STT dropped "warranty"; the agent asked for clarification.
06 refund not received	Refund complaint	**Fail against the reference.** The reference omits the refund complaint the caller made. The identity gate was correct for what was said, so this is a reference mismatch, not an agent error.

Two of six clips pass, and three fail. Speech recognition dropped key terms in clips 04 and 05, and clip 06 is a reference mismatch. Clip 02 has no valid expected row and is not scored. Hinglish return words are missing from the cue list. Latency was not measured.

6. What went wrong

- The first Step 3 default adapter was a hardcoded stub with invented citations. It was replaced with the Step 2 pipeline. An earlier "12/12 pass" was relabelled as fixed-flow only, because it included stub adapters.
- Rule 5 never fired (wrong field names). It was found when a refund case gave a July completion date with no escalation.
- The first batch aborted on a rate-limit error (HTTP 429) at turn 72. Calls are now spaced 4.5 s apart, and the final runs had no quota retries.
- The daily cap of 500 requests was reached during Step 2. One shell timeout may have billed one extra request.
- The first voice runs used a stub. Five clips were sent as turn 1 of new conversations, so those results measured the harness. The clips were re-run with replay.

- Gnani WAV headers carry 0xFFFFFFFF sizes. The runner rewrites only the RIFF and data size fields (`eval/run_voice.py`). A hash check on a copy confirms the data chunk is byte-identical before and after.
- Fix 5 was first applied to text input, and a regex corrupted SKU tokens. It now applies to ASR input only.
- Several agent reports miscounted or mislabelled runs. Each was rechecked against the outputs before use.

7. What would be fixed next

1. Record the composed retrieval query in every transcript. Then search on the customer's question alone and pass order facts only into the prompt. This fixes the PC013 T4 regression and needs a versioned `rag.py`.
2. Compute return deadlines in code (PC003 T4), and include the RRN in escalation replies (PC010 T4).
3. Teach refusal decisions: shipped-order cancellation and late-delivery compensation should be "refuse" (four turns).
4. Voice: confirm key terms when STT misses them, and add Hinglish return words.

8. Bonus work (all in the repo)

- An offline test suite of 59 tests, covering the Step 3 gate, the escalation rules and voice wiring.
- Deterministic refund-date calculation, and an ASR normaliser for SKUs and order IDs.
- A submission checker (`submission/check_submission.py`) and a sample-outputs folder for Steps 2, 3 and voice.